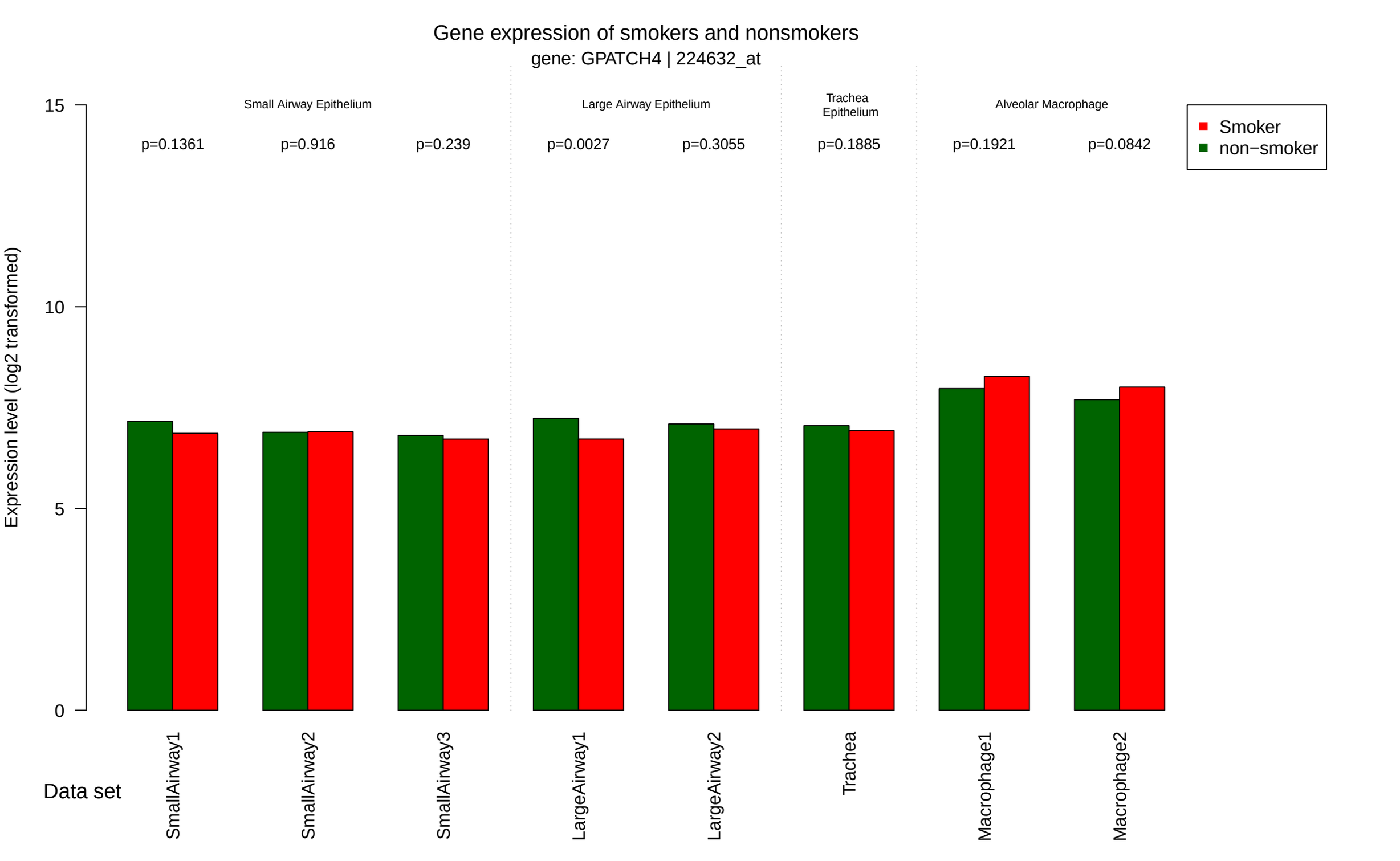
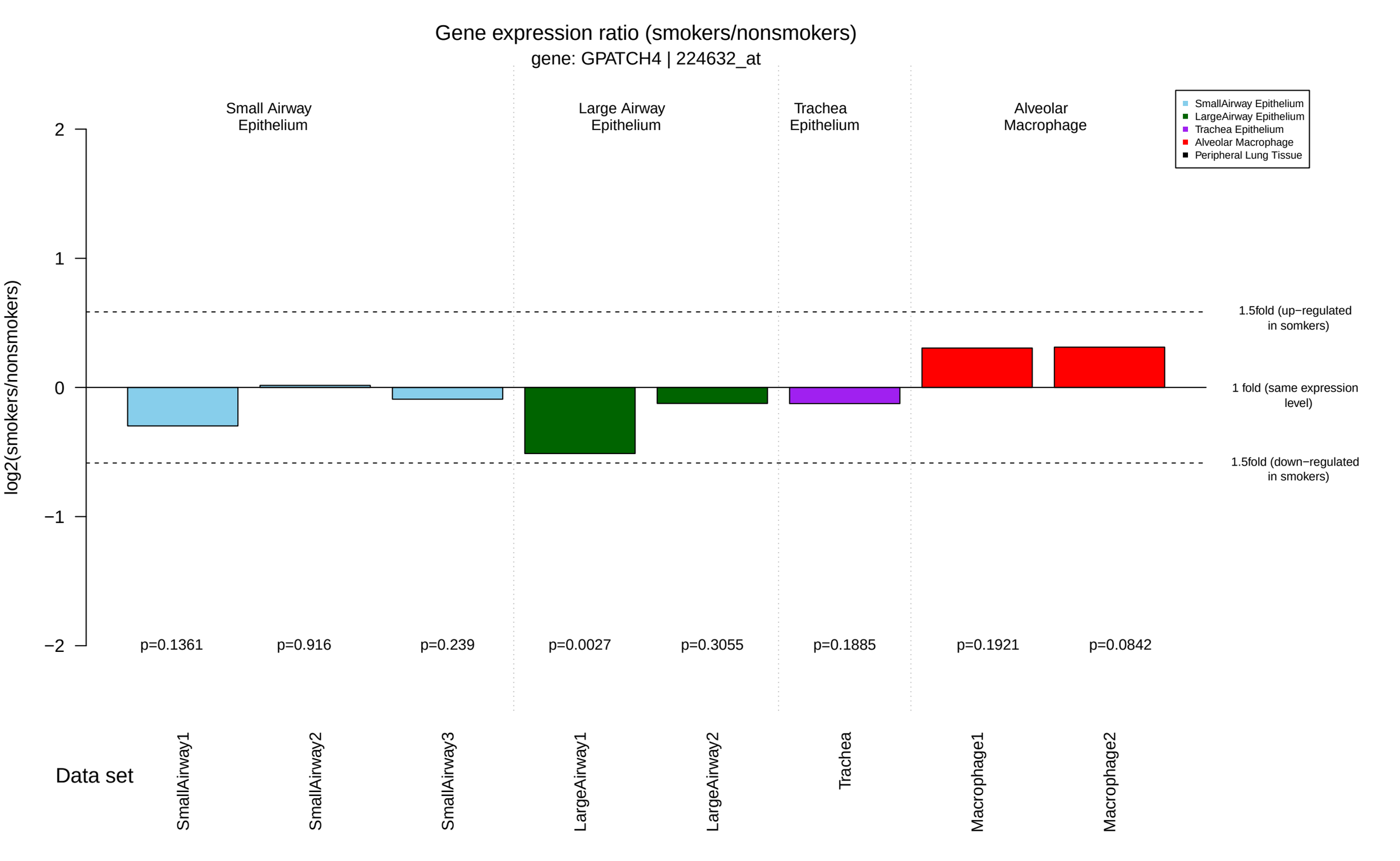
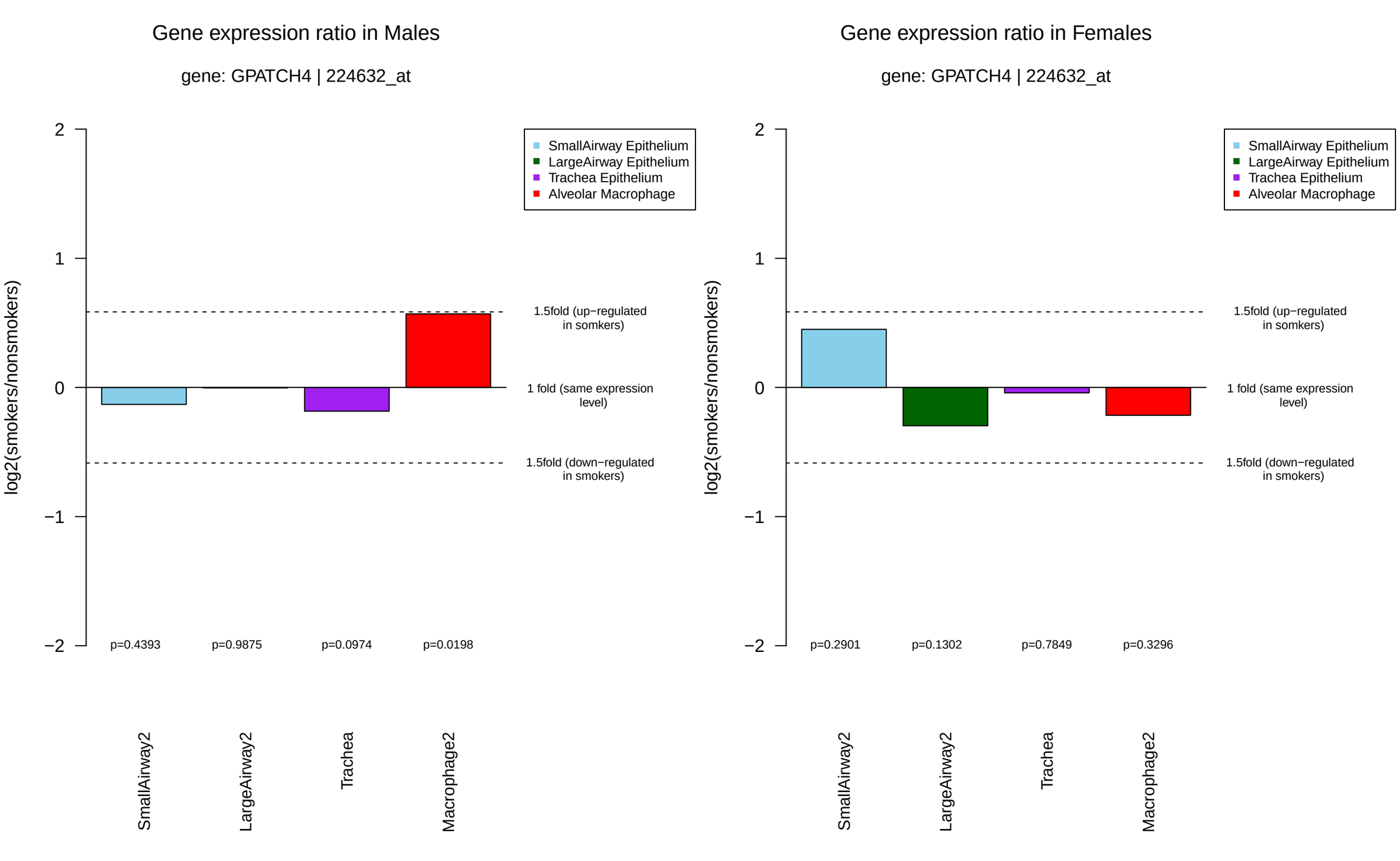
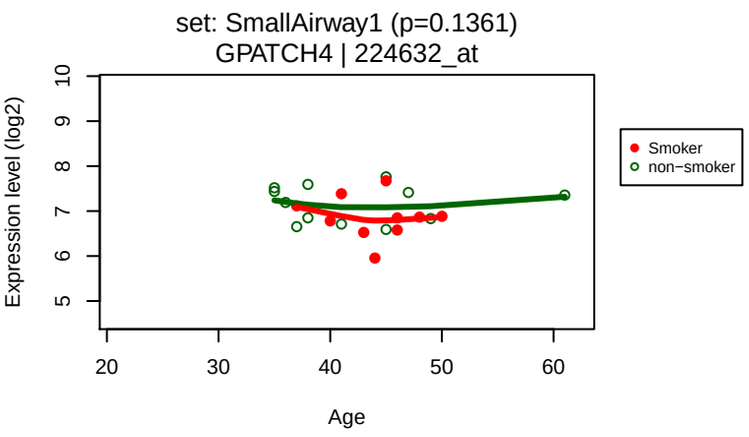
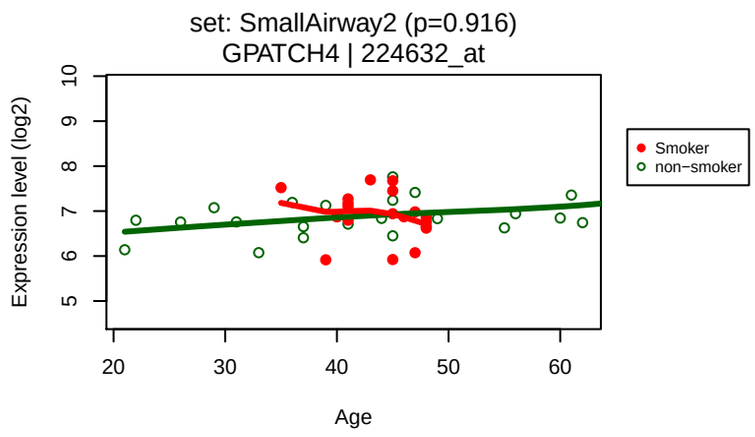
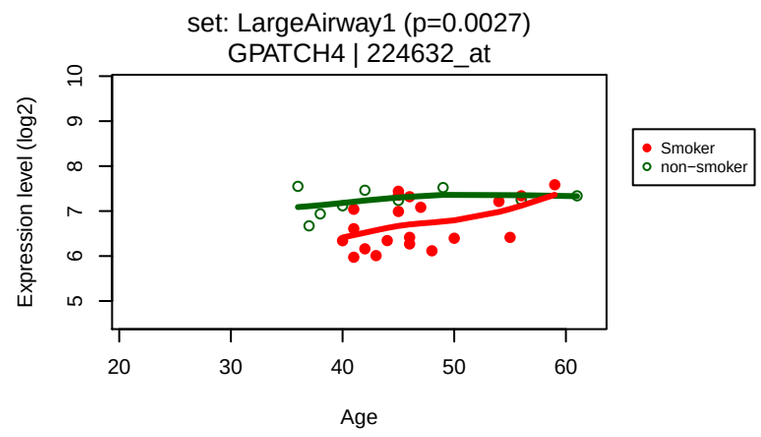

Gene expression of smokers and nonsmokers
gene: GPATCH4 | 224632_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1361
p=0.916
p=0.239
p=0.0027
p=0.3055
p=0.1885
p=0.1921
p=0.0842
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

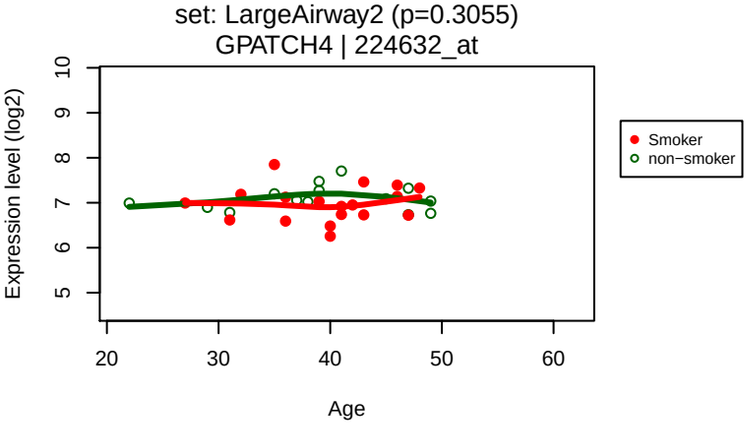

Gene expression ratio (smokers/nonsmokers)
gene: GPATCH4 | 224632_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1361
p=0.916
p=0.239
p=0.0027
p=0.3055
p=0.1885
p=0.1921
p=0.0842
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

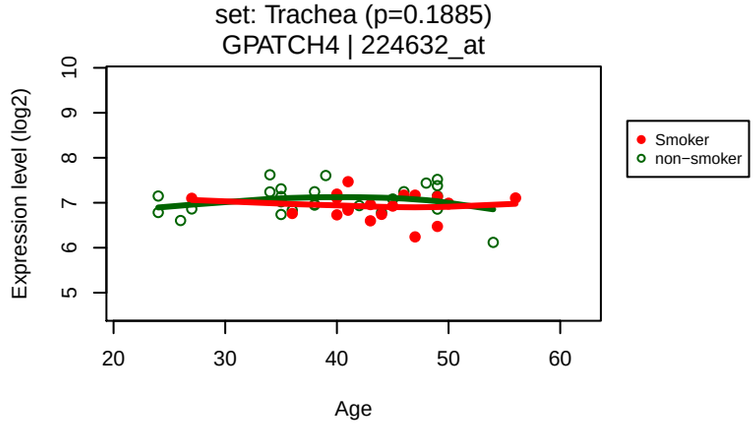

Gene expression ratio in Males
gene: GPATCH4 | 224632_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4393
p=0.9875
p=0.0974
p=0.0198
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPATCH4 | 224632_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2901
p=0.1302
p=0.7849
p=0.3296
SmallAirway2
LargeAirway2
Trachea
Macrophage2

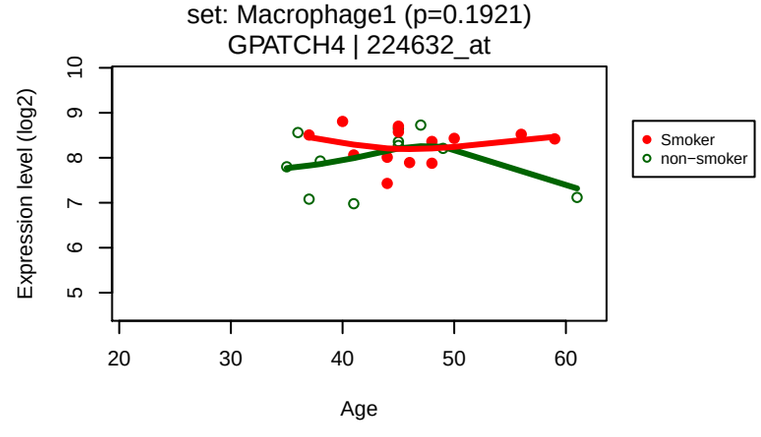

set: SmallAirway1 (p=0.1361)
GPATCH4 | 224632_at
Expression level (log2)
Age
Smoker
non−smoker

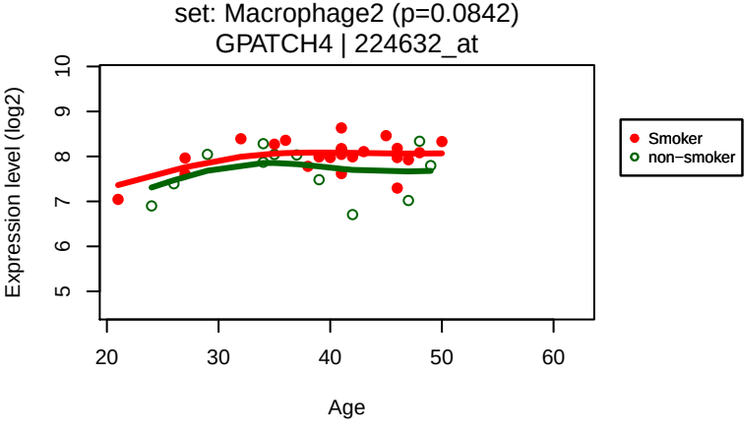

set: SmallAirway2 (p=0.916)
GPATCH4 | 224632_at
Expression level (log2)
Age
Smoker
non−smoker

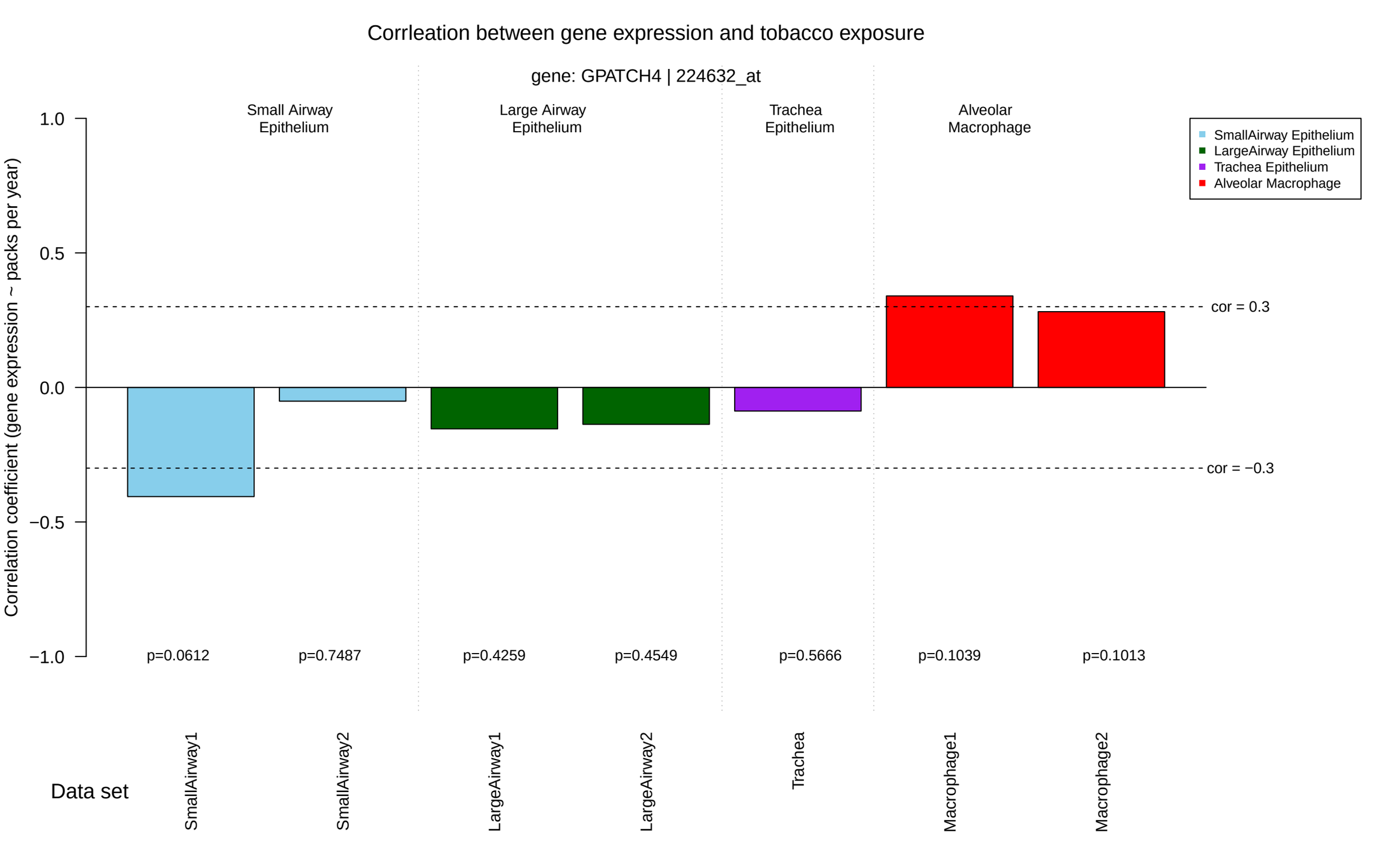

set: LargeAirway1 (p=0.0027)
GPATCH4 | 224632_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3055)
GPATCH4 | 224632_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1885)
GPATCH4 | 224632_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1921)
GPATCH4 | 224632_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0842)
GPATCH4 | 224632_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPATCH4 | 224632_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0612
p=0.7487
p=0.4259
p=0.4549
p=0.5666
p=0.1039
p=0.1013
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2